

Supply Chain Analytics Dashboard Project

Project Overview

The Supply Chain Analytics Dashboard is an interactive Power BI project developed to analyze and monitor supply chain operations, inventory performance, delivery efficiency, supplier risks, and overall business insights. The dashboard helps organizations track important KPIs and make data-driven decisions for improving operational performance.

This project focuses on visual storytelling, business intelligence, and dashboard design using Power BI.

Objective

The main objective of this project is to:

- Monitor supply chain performance in real-time
- Analyze inventory distribution and stock movement
- Track delivery delays and supplier performance
- Identify high-risk suppliers and delayed locations
- Improve decision-making using visual analytics
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Tools & Technologies Used

Tool	Purpose
Power BI	Dashboard creation and visualization
Excel / CSV Dataset	Data source
Power Query	Data cleaning and transformation
DAX	KPI calculations and measures
Data Visualization	Business reporting

Dataset Information

The dataset contains supply chain related information such as:

- Order Details
- Product Categories
- Supplier Information
- Delivery Status
- Revenue & Profit
- Inventory Levels
- Delay Reasons
- Stock Distribution

Dashboard Features

1. Dashboard Overview

This page provides a complete overview of supply chain operations.

KPIs Included:

- Total Orders
- On-Time Delivery Rate
- Delay Rate
- Average Delay
- Suppliers at Risk

Visuals Used:

- KPI Cards
- Donut Charts
- Bar Charts
- Trend Analysis Charts
- Gauge Chart

Insights Generated:

- Highest delayed locations
- Delay distribution by reason
- Monthly order trends
- Delivery performance analysis

2. Inventory Analysis

This section focuses on inventory performance and stock monitoring.

Features:

- Top products by stock
- Stock distribution by location
- Profit category analysis
- Monthly stock trend
- Inventory performance monitoring

Business Benefits:

- Identify overstocked products
- Monitor understock situations
- Improve warehouse management
- Optimize inventory planning

3. Delivery Performance

This dashboard analyzes delivery efficiency and transportation performance.

Metrics Included:

- Delay Percentage
- On-Time Deliveries
- Average Delay Days
- Delivery Trend
- Delay Causes

Business Value:

- Improve delivery management
- Reduce transportation delays
- Monitor logistics performance
- Increase customer satisfaction

4. Supplier Performance

This page evaluates supplier contribution and risk analysis.

Key Analysis:

- Supplier Risk Level
- Supplier Delay Contribution
- Supplier Performance Trend
- Supplier Distribution Analysis

Benefits:

- Identify high-risk suppliers
- Improve supplier relationships
- Monitor supplier efficiency
- Support procurement decisions

Key Insights

- Delay Rate reached 37.5% during the reporting period
- Supplier Issues contributed significantly to delays
- Kolkata and Mumbai recorded the highest delivery delays
- Inventory demand remained stable across months
- On-Time Delivery performance achieved 62.5%
- Profit generation remained positive across product categories

Data Cleaning & Transformation

Data preprocessing was performed using Power Query.

Steps Performed:

- Removed duplicate values
- Handled missing data
- Changed data types
- Renamed columns
- Created calculated measures
- Organized dataset structure

DAX Measures Used

Some important DAX measures used in the project:

Total Orders = COUNT('Supply Chain Data'[Order ID])

Total Revenue = SUM('Supply Chain Data'[Revenue])

Total Profit = SUM('Supply Chain Data'[Revenue]) - SUM('Supply Chain Data'[Unit Cost])

Delay Rate % = DIVIDE([Delayed Orders], [Total Orders])

Dashboard Design Highlights

- Dark themed professional dashboard design
- Interactive visuals and slicers
- Corporate UI layout
- KPI-focused reporting structure
- Easy-to-understand business insights
- Responsive visual arrangement

Conclusion

The Supply Chain Analytics Dashboard successfully provides a centralized view of inventory, supplier, and delivery operations. The project demonstrates how Power BI can transform raw supply chain data into meaningful business insights through interactive dashboards and visual analytics.

This dashboard helps businesses:

- Improve operational efficiency
- Reduce delivery delays
- Optimize inventory planning
- Track supplier performance
- Support strategic decision-making

Future Improvements

Future enhancements for this project may include:

- Real-time data integration
- Predictive analytics
- AI-based forecasting
- Advanced KPI monitoring
- Automated alerts and notifications

Author

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